✂ Control & Coordination — Short Answers

Class X Biology · Every answer trimmed to ≈ 20 words · Print-friendly

PART 1 · Neuron, Synapse & Reflex Arc

Q1. Structure of a neuron — parts (a) & (b). 2 m

Dendrites (a) receive the information; the axon (b) carries it onward as an electrical impulse. Draw and label both. (20 w)

Q2. What is a synapse? Why the delay? 2 m

The synapse is the small gap between neurons. Neurotransmitters diffuse across it, taking time, hence the brief delay. (18 w)

Q3. What happens at the synapse? 2 m

The impulse releases neurotransmitters into the gap; they bind to the next neuron's receptors, generating a new impulse. (18 w)

Q4. Fill (A) & (B) in the reflex arc. 2 m

(A) Sensory neuron carries the impulse to the spinal cord; (B) motor neuron carries it to the effector. (17 w)

Q5. Why processed in spinal cord, not brain? 2 m

Reflexes are processed in the spinal cord for an instant response, avoiding the slower route to and from the brain. (20 w)

Q6. Neuron unit, electrical & chemical signal, backward flow. 3 m

The neuron is the unit. The impulse travels along the axon; its end releases neurotransmitters carried across chemically. Receptors lie only forward, so it is unidirectional. (24 w)

Q7. Reflex action vs involuntary action. 3 m

Reflexes are quick spinal-cord responses to stimuli, like withdrawing from a hot object; involuntary actions are brain-controlled routines, like heartbeat. (20 w)

Q8. Touching a hot object. 3 m

Thermoreceptors in the skin detect heat; the spinal cord integrates; skeletal arm muscles contract to withdraw. (17 w)

PART 2 · Human Brain & CNS Protection

Q9. How is the CNS protected? 2 m

The skull and vertebral column, protective meninges, and cerebrospinal fluid cushion the brain and spinal cord against shocks. (19 w)

Q10. Balance lost after alcohol — which part? 2 m

The cerebellum. It maintains posture and balance and coordinates voluntary muscle movements. (14 w)

Q11. Match brain parts to activities. 3 m

(a) Cerebellum · (b) Medulla · (c) Hypothalamus · (d) Cerebrum · (e) Medulla · (f) Midbrain. (12 w)

Q12. Brain diagram + why medulla injury is fatal. 5 m

Draw the labelled brain. The medulla controls heartbeat, breathing and blood pressure, so injury stops these vital functions immediately. (19 w)

PART 3 · Plant Hormones & Tropic/Nastic Movements

Q13. Phototropism and geotropism. 2 m

Phototropism is growth toward light; geotropism toward gravity. Shoots grow upward (negative), roots downward (positive geotropism). (17 w)

Q14. Why does Mimosa fold on touch? 2 m

On touch, cells in the pulvinus lose water and become flaccid, so leaflets fold through a turgor-pressure change, needing no nerves. (21 w)

Q15. How do pea tendrils climb? 2 m

Tendril cells touching the support grow slower, opposite cells faster, so it coils around — thigmotropism. (16 w)

Q16. Match plant hormones. 3 m

Auxin — cell elongation · Gibberellins — stem elongation and germination · Cytokinins — cell division · ABA — growth inhibitor. (16 w)

Q17. Tropic vs nastic movements. 3 m

Tropic movements are slow, growth-based and directional; nastic movements are fast, non-growth turgor changes independent of stimulus direction. (18 w)

PART 4 · Endocrine System & Feedback

Q18. Why ductless glands? 2 m

Endocrine glands have no ducts; hormones enter the blood directly and reach target organs carrying specific receptors. (16 w)

Q19. Emergency hormone. 2 m

Adrenaline, from the adrenal glands, raises heart rate and blood sugar during fight-or-flight. (13 w)

Q20. Hormone & gland for each condition. 3 m

(a) Goitre — thyroxine, thyroid · (b) Gigantism — growth hormone, pituitary · (c) Diabetes — insulin, pancreas · (d) Delayed puberty — testosterone, testes. (22 w)

Q21. Feedback mechanism — blood sugar. 3 m

High blood sugar triggers insulin, lowering it; low sugar triggers glucagon, raising it — a negative-feedback loop maintaining balance. (19 w)

Q22. Nervous vs endocrine + why hormonal control. 5 m

Nervous control is fast, precise and short-lived; hormonal control is slower but reaches all cells via blood, sustaining long-term growth. (21 w)